

MOLE

Mine Observation & Live-alert Engine

Team WE COOK · Problem SIH26025

Ministry of Coal · Hardware · Disaster Management

Node A + Node B + laptop ML + remote rover + one local dashboard

Tabletop demonstration · 9 September 2026

Idea / prototype document for institute screening and SIH judges

Field	Value
Event	Smart India Hackathon 2026
Official problem ID	SIH26025 (never write SIH2026025)
Problem title	Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India
Organisation	Ministry of Coal
Category	Hardware
Theme	Disaster Management
Team name (portal)	WE COOK
Team ID	To be filled from the SIH portal after nomination
Idea name	MOLE — Mine Observation & Live-alert Engine
Public code	https://github.com/Rithvik-7/WECOOK-MOLE
Document date	9 September 2026
Related portal file	PDF of the official 6-slide idea template only

How to submit

Upload only the official 6-slide idea PDF on the SIH portal. This longer document is for judges, the institute SPOC, and the team. It explains the same product in full. It is not a seventh slide and it is not a substitute for the template PDF.

Honesty, in one paragraph

MOLE is a working tabletop demonstration. Two fixed nodes watch tilt, vibration, and (on Node A) a crack slider. The laptop runs rule-based early warning plus Isolation Forest and a 30-second sensor forecast. An officer can then drive a rover to look more closely. Readings can show disturbance on this model. They do not prove a collapse, name its cause, or certify that a real mine is safe.

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1. Identification

This document describes MOLE as built for Smart India Hackathon 2026. The catalogue ID is SIH26025. That is SIH 2026 problem 025. Judges, catalogues, and the six-slide template match SIH26025. The string SIH2026025 is not the official ID and must not appear on the portal or the idea PPT.

The team name registered for this idea is WE COOK. The GitHub repository name is WECOOK-MOLE. Team ID is issued by the SIH portal after institute nomination and should be written on the title slide when it exists. Member names are not required on the official six-slide PPT. A roster table is in Annex D for institute use.

2. Executive summary

Underground mining can change the ground above it. Officers need a simple way to notice a change, see the evidence, and inspect without walking in first. MOLE answers that need on a tabletop model.

The product is one system, not a pile of parts:

- Node A (fixed): tilt, vibration, and a crack slider.
- Node B (fixed): comparison tilt and vibration. It is not a surveyed geodetic reference.
- USB receiver: Waveshare ESP32-S3-Zero. It only forwards node packets as JSON.
- Laptop ML and rules: Isolation Forest + Local Outlier Factor, a joint A–B forest, and a 30-second sensor-trend forecast. Rules latch the early warning. ML cannot clear a red warning.
- Dashboard: one local website on the laptop. No login. No cloud.
- Rover (core unit): remote drive and inspect on its own Wi-Fi access point. Not autonomous. Not on the node USB radio.

Pitch order for judges: show continuous monitoring and evidence first, then drive the rover because a warning appeared. Do not pitch MOLE as a gas-car maze robot. Do not pitch sensors with a decorative “AI” badge. Machine learning is a core pillar of SIH26025 (AI-enabled and prediction).

3. Official problem statement

Title (as catalogued):

Development of an AI-enabled Low Cost Real Time Mine Subsidence Monitoring, Prediction and Early Warning System for Underground Coal Mines in India

PS word	What the problem asks	What MOLE shows on the table
AI-enabled	Use AI, not only raw sensors	Named sklearn models, features, score, and a written reason from numbers
Low cost	A practical student / field-style kit	ESP32-class boards the team already has; no cloud bill
Real time	Live readings, not a later report	About one summarised packet per second per node; dashboard updates live
Monitoring	Watch the site continuously	Two fixed nodes; rover does not replace them

Prediction	Look a short way ahead	Next 30 seconds of tilt / vibration / crack millimetres, with MAE — not collapse %
Early warning	A clear alarm the officer can act on	WATCH then latched ALERT from rules; AI cannot clear the latch
Underground coal mines	A mine safety context	Ministry of Coal PS; demonstration is a tabletop model, not a certified mine

4. Proposed solution

MOLE helps a mine safety officer notice a concerning change, understand the evidence, and inspect remotely on a tabletop model. Fixed nodes keep watching while the rover, if used, looks in a different place.

Proposed solution · one product, three jobs · rover is not the monitor



Figure 1. Three jobs, one product: monitor, warn and predict, then inspect.

4.1 How it works, in plain steps

- Node A and Node B sample their MPU6050 about 80 times a second and send one summary packet per second over ESP-NOW channel 1.
- The Waveshare ESP32-S3-Zero receives those packets and prints one JSON object per line on USB at 115200 baud. It has no sensors and does not run AI.
- The laptop checks that the packet is usable (fresh, valid ID, valid IMU, Node A slider calibrated).
- The laptop compares tilt with the officer's baseline: $\sqrt{(\text{roll}-\text{roll0})^2 + (\text{pitch}-\text{pitch0})^2}$. Node A slider ADC becomes millimetres after a two-point ruler calibration.
- Rules need three readings in a row. WATCH is 3° or 2 mm. ALERT is 6° or 4 mm and it latches. No packet for 5 seconds is UNKNOWN — never a fake green light.
- Isolation Forest + LOF say whether this combination looks unusual versus this rig's normal. A joint forest watches A versus B. A holdout-selected regression forecasts the next 30 seconds of the same signals and shows MAE.
- The monitoring page explains both the rule and the models. Drive controls stay on a separate rover page so monitoring is not a gamepad.
- If the officer decides to inspect, the laptop joins Mine-Rover-AP and proxies hold-to-move commands. The rover reports distance, an IR flag, MQ-7 raw ADC, and rover IMU. Rover vibration is never treated as Node B.

4.2 What we do not claim

Allowed on this demo	Not allowed
NORMAL / WATCH / ALERT / UNKNOWN on this rig	"The mine is safe / certified / collapse imminent"

Isolation Forest score and top feature versus this tabletop's normal	Collapse or subsidence probability
30 s forecast of tilt / vibration / crack mm + MAE	"Roof will fail in N minutes" as a geotech claim
MQ-7 raw ADC and a local NORMAL/HIGH flag	CO ppm from an uncalibrated MQ-7
IR digital flag (LOW = near). It does not stop motors	Collision avoidance, lidar, or auto-brake
Remote FWD / REV / LEFT / RIGHT / STOP	Autonomous navigation
Operator-driven inspection	Camera vision, LLM geology, login, cloud, or LoRa

5. Innovation and uniqueness

- One product, two radios. Nodes use ESP-NOW into USB. The rover uses its own Wi-Fi AP. Mixing those paths would poison the comparison story and the ML features.
- Rules and ML sit side by side. The latch is independent. Isolation Forest cannot talk a red warning back to green.
- Prediction is honest. The dashboard label is a 30-second sensor-trend forecast, not a collapse prediction. MAE is visible.
- Node B is a comparison location, not a fake surveyed monument. Joint Isolation Forest uses |A-B| tilt and vibration, not a geology model.
- Green means fresh valid data and no configured tabletop trigger. It does not certify a mine.
- Simulation is permanently labelled and cannot persist-train Isolation Forest. Live USB never silently fakes data.

6. System architecture

Intelligence is split on purpose. ESP32 boards measure and send. The laptop calibrates, stores, compares A with B, runs rules and ML, and draws the website. Rover driving is remote, not autonomous.

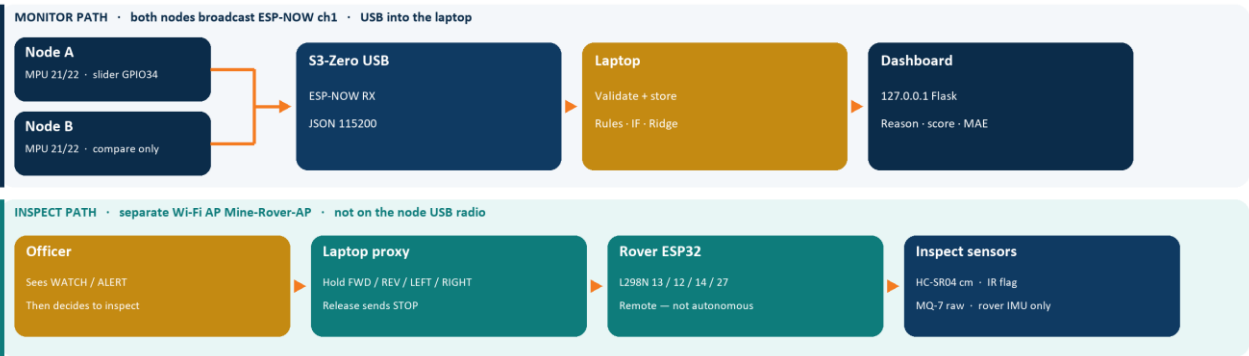


Figure 2. Monitor path (ESP-NOW → S3-Zero USB → laptop) is separate from inspect path (Mine-Rover-AP).

7. Hardware

The team already has the parts. This freeze is the as-built contract. Do not invent a different board, pin, or radio for the pitch.

Unit	MCU	Job	Radio
------	-----	-----	-------

Node A	Classic ESP32-WROOM-32 DevKit	Tilt + vibration + crack slider	ESP-NOW channel 1 broadcast
Node B	Same DevKit	Comparison tilt + vibration, no slider	ESP-NOW channel 1 broadcast
Receiver	Waveshare ESP32-S3-Zero (ESP32-S3FH4R2)	ESP-NOW → USB-C JSON 115200. No sensors	STA, channel 1
Rover	Classic ESP32-WROOM-32 DevKit	Drive + inspect	Wi-Fi AP Mine-Rover-AP only

The receiver is not a classic ESP32 and not an ESP32-S2. In Arduino IDE choose Waveshare ESP32-S3-Zero or ESP32S3 Dev Module, and set USB CDC On Boot = Enabled. Native USB, not a CH340 adapter.

7.1 Node A and Node B

Same sketch family: firmware/node/node.ino (A) and firmware/node_b/node.ino (B). Firmware does not compute millimetres, alerts, or AI.

Function	Detail
MPU6050	I ² C 0x68, SDA GPIO21, SCL GPIO22, 3.3 V, AD0 to GND
Node A slider	10 kΩ linear pot, SIG → 1 kΩ → GPIO34 (ADC1). Never 5 V into the ADC. NODE_ID=1, HAS_POT=1
Node B slider	Not fitted. sliderRaw = -1. NODE_ID=2, HAS_POT=0
Sample rate	MPU about 80 Hz (SAMPLE_US 12500); one summarised packet per second
Broadcast	ESP-NOW FF:FF:FF:FF:FF:FF on channel 1. Unencrypted. Tabletop range only
valid bitmask	IMU=1, potentiometer=2, vibration window=4. Healthy A=7, healthy B=5. Invalid fields are JSON null, never 0
Tilt on the laptop	$\sqrt{(\text{roll}-\text{roll0})^2 + (\text{pitch}-\text{pitch0})^2}$ after the officer captures a baseline
Orientation in firmware	Gravity roll/pitch. Trustworthy only when the mount is approximately still
vibration_g	RMS residual around the one-second mean. Not a calibrated seismic measure

7.2 USB receiver

- Sketch: firmware/receiver_s3/receiver_s3.ino.
- Wi-Fi STA forced to channel 1, ESP-NOW receive, size / version / node_id check.
- Prints one JSON object per line on USB Serial 115200. May add gateway_ms.
- Does not compute tilt-delta, millimetres, rules, or AI.
- If upload fails: hold BOOT, tap RESET, release BOOT, then upload.

7.3 Rover (core unit; radio stays separate)

Sketch: firmware/rover/rover.ino. Access point SSID Mine-Rover-AP, IP 192.168.4.1, open AP in firmware (no password). Leave L298N ENA/ENB jumpers ON. Firmware stops by setting IN1–IN4 LOW, plus a 400 ms watchdog if no repeated command. GPIO19 is the IR flag, not a motor pin.

Function	Pin / note
L298N IN1 / IN2 (left)	GPIO13 / GPIO12
L298N IN3 / IN4 (right)	GPIO14 / GPIO27
MPU6050	SDA 21 / SCL 22 / 0x68 / 3.3 V
HC-SR04 TRIG	GPIO5
HC-SR04 ECHO	GPIO18 after a 1 k Ω / 2 k Ω divider. ECHO is 5 V — never wire straight in
MQ-7 analog	GPIO36, raw 12-bit ADC. Local HIGH if raw $\geq$ 2500. Not ppm
IR OUT	GPIO19, INPUT_PULLUP, LOW = obstacle, majority-of-3 debounce. Does not stop motors

Command	IN1 IN2 IN3 IN4
STOP	0 0 0 0
FWD	1 0 1 0
REV	0 1 0 1
LEFT	0 1 1 0
RIGHT	1 0 0 1

If a side runs backward, swap that motor's two wires. Do not invent new GPIOs. Rover HTTP GET: /forward /backward /left /right /stop /telemetry /.

8. Communication design

Node path and rover path must stay separate. The S3-Zero USB path does not carry rover commands or video. The laptop must join Mine-Rover-AP to drive. Live USB drop is retried every 2 s; nodes become UNKNOWN after 5 s without a valid packet. Live mode never falls back to simulation.

Path	Medium	Payload	Must not
Nodes → laptop	ESP-NOW ch1 → S3-Zero → USB JSON 115200	Node A/B summaries	Carry rover drive or video
Laptop → rover	Wi-Fi AP Mine-Rover-AP, HTTP	Drive commands + inspect telemetry	Enter node ML features

9. Software

Stack: Python 3, Flask, SQLite, scikit-learn on the laptop only, pyserial, joblib, pytest. No accounts, no npm, no CDN required. Charts are canvas, not Chart.js from the internet. Server binds 127.0.0.1:5000.

Module	Responsibility
app.py	Flask app, USB thread, rover poll thread, HTTP routes
engine.py	Lock, ingest, freshness, baselines, snapshot JSON, forecast cache
models.py	Telemetry / RoverTelemetry, schema 1 parse
serial_reader.py	COM port, readline JSON, auto / COMx

simulate.py	Node scenarios + SimulatedRover
database.py	SQLite packets/events, CSV export tagged live/sim
calibration.py	Two-point ADC → mm
rules.py	WATCH / ALERT / UNKNOWN, persist 3, latch, ack ≠ clear
features.py	Per-node vectors + joint residual vector
anomaly.py	Per-node Isolation Forest + LOF; prior/live joblib
forecast.py	Ridge / Huber / LinearRegression vs persistence; 30-point horizon
explain.py	Template reasons from numbers (not an LLM)
rover_client.py	HTTP to http://192.168.4.1
rover_sense.py	Honest IR vs ultrasonic copy (stops_motors: false)
train_prior.py	Shipped tabletop *.prior.joblib files
helper.py	Optional local FAQ / Ollama helper named Pip. Not the product mine-AI

Pages: Monitoring + AI at /monitoring (also /). Rover Inspection at /rover. Drive controls must not appear on the monitoring page. Simulate shows a permanent Rehearsal bar and cannot POST /api/train. Tabletop prior models still load so Isolation Forest can be READY without waiting 120 live samples. Those priors describe a quiet tabletop, not a real mine, and the UI labels them as a tabletop prior.

Pip, if shown, answers from a local handbook and FAQ. Product AI remains sklearn. A helper chatbot is not the SIH “AI-enabled” claim.

9.1 HTTP API (laptop)

Method	Path	Notes
GET	/ /monitoring /rover	HTML pages
GET	/api/state	Full monitoring snapshot
GET	/api/export.csv	History, tagged live or sim
POST	/api/baseline	{node_id: 1 or 2}
POST	/api/ack	Awareness only; does not erase history
POST	/api/clear	Needs 3 recovered NORMAL samples; 409 if not
POST	/api/train	Live only; ≥120 NORMAL; blocked if WATCH/ALERT/UNKNOWN/sim
POST	/api/calibration	adc0, adc1, mm0, mm1 — distinct ADC
POST	/api/scenario	Simulate only: normal, rising, watch, alert, offline, sensor_fault
POST	/api/rover/<cmd>	forward, backward, left, right, stop
GET	/api/rover/state	Rover telemetry + IR explanation

10. Early-warning rules

These numbers are tabletop demo thresholds. They are not DGMS trigger levels and they are not a scientific mine rule.

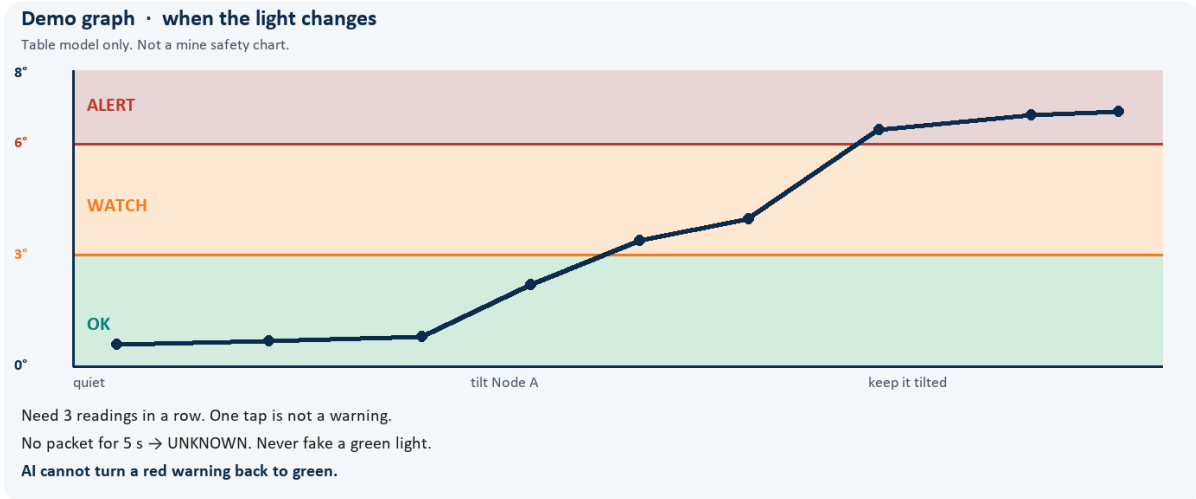


Figure 3. Demo tilt bands used on the table. Not a mine safety chart.

Rule	Tabletop setting
WATCH	Tilt $\geq 3^\circ$ or crack ≥ 2 mm, 3 consecutive samples
ALERT	Tilt $\geq 6^\circ$ or crack ≥ 4 mm, 3 consecutive samples; ALERT latches
UNKNOWN	Stale > 5 s, unseen node, invalid IMU, or Node A slider not millimetre-calibrated
Ack	Records that the officer saw it. Does not clear the latch
Clear	Needs 3 fresh NORMAL samples
Independence	Node B never cancels Node A. Isolation Forest cannot clear a rule alert

11. Machine learning (core product, laptop only)

SIH26025 asks for AI-enabled monitoring and prediction. MOLE puts that on the laptop with scikit-learn. There is no camera vision model and no LLM geology. Rover IMU, IR, ultrasonic, and MQ-7 are not ML features.

11.1 Isolation Forest + Local Outlier Factor

- sklearn IsolationForest (contamination 0.05) + LocalOutlierFactor, with StandardScaler.
- Features: tilt_change_deg, vibration_g, d_tilt, and Node A relative_mm once calibrated.
- States: INACTIVE / READY / UNUSUAL. The officer sees score, LOF vote, top standardised feature, feature names, and train count.
- Optional live retrain writes nodeN.joblib after ≥ 120 NORMAL samples in live mode. Simulate cannot persist-train.
- Shipped artifacts/node1.prior.joblib, node2.prior.joblib, and joint.prior.joblib describe quiet-tabletop NORMAL so a demo can start READY.

11.2 Joint A–B forest

A third Isolation Forest uses $[\text{abs}(A_{\text{tilt}} - B_{\text{tilt}}), \text{abs}(A_{\text{vib}} - B_{\text{vib}})]$. Patterns such as QUIET / LOCAL_A / LOCAL_B / COMMON help the officer see local versus common motion. This is not a geology model.

11.3 Thirty-second sensor forecast (the hackathon “prediction”)

- Window of recent ~1 Hz samples. Candidates: Ridge, HuberRegressor, LinearRegression, versus a persistence baseline.
- Holdout MAE picks the winner. Horizon is the next 30 samples of tilt, vibration, and Node A gap.
- UI: solid measured line, dashed forecast, error band, MAE, quality, surprise z. Time-to-WATCH/ALERT only if the fitted slope heads toward the threshold.
- Quiet data can let LinearRegression win with MAE near 0. That is a flat signal, not magic. Huber may fail to converge; it is a candidate, not sacred.
- On-screen label: 30 s sensor trend forecast — not a collapse prediction.

12. Rover inspection

The rover exists so the officer can look after a warning, not so the product becomes a maze robot. Hold-to-move; release or Space sends STOP. The laptop sends STOP twice to beat in-flight races. Firmware also stops after about 400 ms.

Sensor	What the officer sees	Limit
HC-SR04	Distance ahead in centimetres	Hobby ultrasonic, not lidar
IR	Digital near flag	Not a range; does not auto-brake; can disagree with ultrasonic
MQ-7	Raw ADC and local NORMAL/HIGH	Not carbon monoxide ppm
Rover MPU	Roll / pitch / vibration on the rover	Never merged into Node B or node ML

In simulation, IR can trip under about 12 cm while ultrasonic “tight” is 25 cm, so they can disagree on purpose. On live hardware the pins are independent.

13. Operator interface

- Monitoring + AI: node cards with tilt from baseline, rule status and reason, Isolation Forest evidence, forecast chart with MAE, joint A–B note, event history, CSV export.
- Rover Inspection: hold FWD/REV/LEFT/RIGHT, STOP, centimetres, IR explanation, MQ-7 raw, rover IMU, connection status.
- Train A/B buttons are disabled in simulate.
- Rehearsal “Rising trend” climbs Node A while Node B stays quieter — a judge-friendly forecast and localisation demo.

How we show it works (one loop on the table)

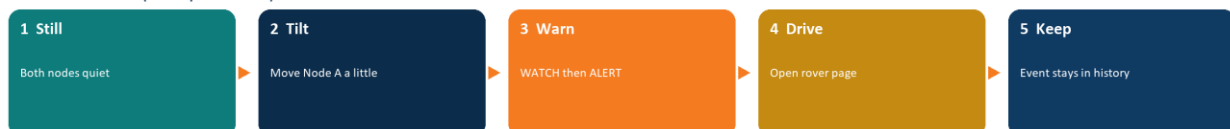


Figure 4. One demonstration loop on the table.

14. Data honesty and safety of claims

Green / NORMAL means fresh valid data and no configured tabletop trigger. It does not certify a mine. Tilt measures orientation change from a captured baseline. It is not vertical ground settlement in centimetres. Hobby ESP32 / MPU6050 electronics are not intrinsically safe and are not DGMS-approved underground

instruments. ESP-NOW on this model is unencrypted and short-range. Do not claim LoRa, mesh, or colliery networking.

15. Feasibility and viability

The idea is feasible as a Smart India Hackathon Hardware demonstration this week: boards are with the team, firmware sketches and the local website are in the repository, and simulation works with no hardware. Mine certification, intrinsically safe equipment, and surveyed geodetic mounts are out of scope.

Check	Status
Boards for A, B, rover, S3-Zero	Available with the team
Firmware	In repo (flashing on hardware day still required)
Laptop website	Flask, local, no login, no cloud
ML without a long wait	Tabletop priors load READY; optional live retrain later
Demo without boards	--mode simulate, permanently labelled
Radio	ESP-NOW tabletop; rover AP across the table, not through rock
If USB dies	UNKNOWN. Never fake green
If rover Wi-Fi dies	Motors stop in about 0.4 s

16. Risks and how we handle them

Risk	Why it is real	What we do
Over-claiming the PS	Tilt is not settlement in cm; Isolation Forest is not collapse probability	Honesty table on the dashboard and in this document
Mixing rover with nodes	A moving IMU would poison Node B and node ML	device_id rover; never in node models
False comfort from a dead sensor	Silence can look “fine”	Stale >5 s → UNKNOWN; ML cannot clear a latch
Uncalibrated slider	ADC is not millimetres until two-point cal	Node A cannot show NORMAL until calibrated
IR disagrees with ultrasonic	Cheap IR is a flag, not a tape measure	Dashboard explains; IR does not cut motors
Hobby electronics underground	Not approved for a real mine atmosphere	Table model only. Field needs approved gear

17. Impact and benefits

The user on demo day is a mine safety officer on a tabletop model. The Ministry of Coal owns the problem statement. Impact we can defend is notice + evidence + a remote look — not disaster-prevention numbers we cannot prove.

Benefit a judge can verify	How	Not a benefit we invent
See A vs B at the same time	Two node cards, joint forest	A real mine network

See why a warning latched	Rule reason in words from numbers	Certified early warning
See AI evidence	Model names, features, score, MAE	Collapse %
Inspect without entering the model tunnel first	Open rover page after a warning	Autonomous rescue robot
Keep a record	Ack, history, CSV	Erasing the event by clicking Ack
Run without venue internet	127.0.0.1 Flask; USB nodes still work if Wi-Fi is bad	Cloud operations centre

18. Demonstration protocol for judges

- Start simulate or live. Show both nodes updating. State SIH26025: monitor → warn → inspect.
- Show model names, train count or “Tabletop prior”, features, forecast MAE. Say it is not collapse %.
- Use Rising trend, or physically tilt Node A. Keep Node B quieter. That is localised evidence.
- WATCH then ALERT with the measured reason. Rules and ML side by side.
- Open Rover Inspection because a warning appeared.
- Raise wheels first. Hold FWD, then STOP. Show centimetres, IR flag, MQ-7 raw. IR does not auto-brake.
- Ack. Recover the mounts. Clear after three NORMAL samples. History still lists the event.

Spoken close: Isolation Forest + LOF flag unusual combinations versus this rig’s learned normal. A joint forest watches A vs B residual. Holdout-selected sklearn regression forecasts the next 30 seconds of those same signals with MAE. Rules remain the independent early-warning latch. The rover is remote inspection, not autonomy.

19. Implementation status (9 September 2026)

Done in this repository	Still required on hardware day
Firmware sketches for four boards	Wire to the frozen pin map (especially echo divider and GPIO19 = IR)
Flask two-page website	Flash four sketches in Arduino IDE
Rules, IF+LOF+joint, holdout forecast, priors	Serial Monitor on S3-Zero: JSON for node_id 1 and 2 at 115200
Simulate scenarios, rover proxy, IR honesty, CSV	python app.py --mode live --serial auto (or COMx)
Contract tests (25) in test_app.py and test_complete.py	Baseline A/B + slider two-point calibration
Official 6-slide idea PPT + this document	Join Mine-Rover-AP, polarity test, STOP watchdog, IR/ultrasonic sanity

SIH26025 is a Hardware problem. Extra sklearn does not replace working boards. Firmware exists in the repo but was not necessarily already flashed on the team’s boards when this document was written.

20. How to run the prototype

Without hardware (rehearsal):

cd software

python -m pip install -r requirements.txt

python app.py --mode simulate

Then open <http://127.0.0.1:5000/monitoring> and <http://127.0.0.1:5000/rover>

With hardware:

- Flash Node A, Node B, S3-Zero (USB CDC On Boot Enabled), and rover.
- Confirm JSON for node_id 1 and 2 in Serial Monitor at 115200. Close Serial Monitor before starting Python.
- `python app.py --mode live --serial auto` (or `--serial COM5`).
- Capture Baseline A and Baseline B while still. Calibrate the Node A slider with two ruler points.
- Join Mine-Rover-AP. Raise wheels. Test FWD/STOP. Confirm centimetres, IR, MQ-7 raw.

`python -m pytest -q` from the software folder runs the contract tests.

21. Research and references

Only items we actually use, plus published mine-safety context labelled as context only. No invented accident statistics.

Item	Why it is here	Citation / link
PS SIH26025	Official catalogue ID	SIH 2026 portal · Ministry of Coal · Hardware · Disaster Management
Isolation Forest	Anomaly versus this rig's normal	Liu, Ting, Zhou — Isolation Forest, IEEE ICDM 2008
30 s forecast	Short-horizon prediction of measured signals	scikit-learn Ridge / Huber / LinearRegression; holdout MAE
MPU6050	Gravity tilt + RMS vibration proxy	InvenSense MPU-6000/6050 Product Specification
ESP-NOW + ADC1	Node radio ch1; slider on GPIO34	Espressif ESP-NOW User Guide; ESP32 ADC notes (ADC1, not ADC2 with Wi-Fi)
ESP32-S3-Zero	USB CDC receiver, no sensors	Waveshare ESP32-S3-Zero wiki
MQ-7 / IR	Rover inspect only	MQ-7 datasheet — raw ADC, not ppm. IR = digital LOW flag
Implementation	Working software + firmware in this repo	https://github.com/Rithvik-7/WECOOK-MOLE
Domain context only	Why mines care about movement	Coal Mines Regulations, 2017; DGMS (S&T) Circ. 01/2017 cl. 6.1 — not implemented here

22. Glossary

Term	Meaning in MOLE
MOLE	Mine Observation & Live-alert Engine — the whole system
Node A / Node B	Fixed tabletop sensors. A has the slider; B is comparison only
S3-Zero	Waveshare ESP32-S3 USB receiver. Not a monitoring node
Latch	ALERT stays until a proper clear; ML cannot cancel it

MAE	Mean absolute error of the 30 s signal forecast
Tabletop prior	Shipped quiet-table joblib. Not a real-mine model
UNKNOWN	We do not know; never shown as healthy green
Mine-Rover-AP	Rover Wi-Fi name. Separate from node ESP-NOW

Annex A — Short text for the SIH portal idea box

MOLE is a tabletop mine-monitoring and inspection system for SIH26025. Two fixed ESP32 nodes measure tilt, vibration, and (on Node A) a crack slider, and send ESP-NOW to a Waveshare ESP32-S3-Zero USB receiver. A local laptop website stores every reading, latches WATCH/ALERT from simple rules, and runs Isolation Forest plus a 30-second sensor-trend forecast with MAE. After a warning, the officer remotely drives a separate rover on Wi-Fi AP Mine-Rover-AP to inspect. Measurements show disturbance on this model. They do not prove a collapse or certify a mine. No cloud, login, camera vision, LoRa, CO ppm, or autonomous driving.

Annex B — Node packet and USB JSON

On air (packed C struct):

version (1), nodeId (1 or 2), sequence, uptimeMs, roll, pitch, vibration, sliderRaw (Node B always -1), flags (IMU=1, POT=2, VIB=4).

USB example, Node A:

```
{"type":"telemetry","schema":1,"node_id":1,"seq":17,"uptime_ms":17000,"gateway_ms":18500,"valid":7,"roll_deg":0.12,"pitch_deg":-0.08,"vibration_g":0.004,"adc_raw":2048}
```

USB example, Node B:

```
{"type":"telemetry","schema":1,"node_id":2,"seq":17,"uptime_ms":17000,"gateway_ms":18500,"valid":5,"roll_deg":0.03,"pitch_deg":0.06,"vibration_g":0.003,"adc_raw":null}
```

node_id 1 and 2 are reserved. Rover uses device_id "rover" over HTTP, not this USB schema.

Annex C — Flash map

Unit	Sketch	Arduino board setting
Node A	firmware/node/node.ino	ESP32 Dev Module; NODE_ID 1, HAS_POT 1
Node B	firmware/node_b/node.ino	ESP32 Dev Module; NODE_ID 2, HAS_POT 0
Receiver	firmware/receiver_s3/receiver_s3.ino	Waveshare ESP32-S3-Zero or ESP32S3 Dev Module; USB CDC On Boot = Enabled
Rover	firmware/rover/rover.ino	ESP32 Dev Module; AP Mine-Rover-AP

Annex D — Team roster (institute use)

Official six-slide PPT does not require a member table. Fill this for college nomination if the SPOC asks. Do not invent names on the idea PPT.

#	Name	Suggested role on this build
1		Node A wiring, mount, slider calibration

2		Node B and tabletop mechanics (movable surface)
3		S3-Zero USB receiver and node radio
4		Rover mechanics, motors, inspect payload
5		Laptop ingest, rules, storage, rover proxy
6		Dashboard, ML evidence, demo script, pitch

Team ID (portal): _____ Institute SPOC: _____

Annex E — Related files in this repository

- Official idea PPT/PDF: pitch/SIH26025_WE_COOK_IDEA.pptx and .pdf (upload the PDF of 6 slides).
- This document: pitch/SIH26025_WE_COOK_MOLE_Judge_Document.docx / .pdf.
- Speaker crib for the 6 slides: pitch/SPEAKER-CRIB.md.
- As-built contract: AGENTS.md.
- How to run: README.md.
- Firmware: firmware/README.md. Telemetry examples: software/TELEMETRY.md.

Close: nodes detect → rules latch → Isolation Forest + Ridge explain → rover inspects. Team WE COOK · SIH26025 · tabletop only.